

Debajyoti Deb

PhD Student, Civil & Environmental Engineering
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EDUCATION

- Louisiana State University** Baton Rouge, Louisiana, USA
Doctor of Philosophy in Civil & Environmental Engineering August 2023 - Expected 2028
 - Research Area:** Biological waste treatment — microbial treatment of wastewater
 - Advisor:** Dr. William M. Moe, Marvin Rex Clemmons Professor in the College of Engineering
- Bangladesh University of Engineering and Technology (BUET)** Dhaka, Bangladesh
Bachelor of Science in Chemical Engineering February 2017 - May 2022

PUBLICATIONS

- W. M. Moe, L. Albuquerque, **D. Deb** (2025). Complete genome sequence of the type strain *Ampullimonas aquatilis* LMG 28208^T. *Microbiology Resource Announcements* 14(8), e00384-25. doi:10.1128/mra.00384-25
- D. Deb**, W. M. Moe (2025). Complete genome sequence of the type strain *Piscinibacterium candidicorallinum* LMG 29480. *Microbiology Resource Announcements* 14(5), e00139-25. doi:10.1128/mra.00139-25

RESEARCH EXPERIENCE

- Louisiana State University** Baton Rouge, Louisiana, USA
Graduate Research Assistant August 2023 - Present
 - Focus:** Doctoral research in biological waste treatment under Dr. William M. Moe, on the microorganisms and microbial processes involved in wastewater treatment, nutrient transformation and contaminant biodegradation
 - Cultivation:** Preparing cultures and subcultures for the isolation and maintenance of bacterial strains under aseptic conditions; microbial enumeration by most probable number (MPN)
 - Molecular Characterisation:** DNA extraction, PCR, qPCR and gel electrophoresis; 16S rRNA sequence inspection and alignment in BioEdit
 - Taxonomic Characterisation:** Gram staining, light microscopy, and physiological and biochemical assays used to describe and classify bacterial strains
 - Genomics:** Genome annotation (RAST) and comparative genomic analyses to support bacterial classification and genotype–phenotype interpretation
 - Output:** Contributed to two complete reference genome studies of environmentally relevant bacterial type strains; first author on one
 - Funding:** Genome studies supported by the Governor’s Biotechnology Initiative of the Louisiana Board of Regents, through the LSU Hazardous Substance Research Center Environmental Biotechnology Initiative

RESEARCH INTEREST

- Biological Waste Treatment
- Bioremediation
- Bacterial Characterization
- Environmental Microbiology
- Microbial Genomics
- Environmental Biotechnology

SKILLSET

- Microbiology:** Bacterial culturing and subculturing, aseptic technique, isolation and maintenance of strains, microbial enumeration (MPN), Gram staining, light microscopy, autoclaving, taxonomic characterisation assays
- Molecular & Genomic:** DNA extraction, PCR, qPCR, gel electrophoresis, 16S rRNA sequence analysis, genome annotation (RAST), comparative genomics
- Software:** BioEdit (sequence alignment and editing), Aspen HYSYS, MATLAB, AutoCAD (2D), Microsoft Office
- Languages:** Bangla (native), English (full professional proficiency)

GRADUATE COURSEWORK

- Advanced Topics in Biodegradation
- Theory and Operation of Wastewater Treatment Facilities
- Physical/Chemical Processes in Water and Wastewater Treatment
- Advanced Topics in Water Quality and Treatment
- Design of Solid and Hazardous Waste Management Systems
- Special Topics in Environmental Engineering Science
- Special Topics in Biological Engineering
- Sustainability Engineering
- Integrated Environmental Issues
- Environmental Planning and Management
- Applied Ecology
- Experimental Statistics I

INDUSTRY EXPERIENCE

- **British American Tobacco Bangladesh** Dhaka, Bangladesh
Team Leader - Secondary Manufacturing Department October 2022 - November 2022
 - **Operations:** Coordinated equipment owners and supported daily shift execution; monitored production and shift performance
 - **Safety & Quality:** Supported safety and quality compliance; helped troubleshoot equipment and process issues with the operations team
- **ASM Chemical Industries Limited** Gazipur, Bangladesh
Industrial Attachment Trainee May 2022
 - **Training:** Two-week attachment with a chlor-alkali manufacturer producing stable bleaching powder, poly-aluminium chloride, caustic soda and chlorinated paraffin wax; plant operations, utilities, process-safety systems, SOPs and QC lab testing

LEADERSHIP

- **Bangladeshi Students' Association (BSA) at LSU** Baton Rouge, Louisiana, USA
Program Coordinator August 2025 - May 2026
 - **Concert:** Led logistics for the first-ever Bangladeshi rock band concert in Louisiana (November 2025), drawing 300+ attendees including visitors from nearby states; planned ticket sales, entry flow, and crowd and venue management
 - **Operations:** Coordinated performer arrival and escort, directed a 12-person volunteer team, and kept the program on schedule
 - **Recognition:** Co-led 16+ BSA programs in 2025–26; BSA was one of four Gold Organizations at LSU's 2026 Tiger Excellence Awards

EARLIER RESEARCH & UNDERGRADUATE PROJECTS

- **Bio & Environmental Technology Research Group, BUET** Dhaka, Bangladesh
Undergraduate Research Assistant May 2018 - March 2020
 - **Bio-Fuel:** Extraction of ethanol from *Chlorella vulgaris*; scaled algal cultures from 200 mL to 25 L by tuning culture composition, temperature and humidity
 - **Undergraduate Thesis:** Cleaner production practice in the power generation sector of Bangladesh — process development, hazard reduction and waste minimisation across plants of several generation technologies
 - **Undergraduate Design:** 100 tons per day plant re-refining used industrial lube oil into API Group II base oil by thermal de-asphalting

TEACHING EXPERIENCE

- **Instructor:** UDVASH and OMECA coaching institutes, Bangladesh — classroom instruction in mathematics, thermodynamics, reaction kinetics and electrochemistry for university admission candidates (2018 - 2022)
- **Content Development:** Produced recorded lecture material for Emerging Study, an ed-tech platform